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# HAOLIN YANG

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## EDUCATION

### University of Chicago

Sept. 2024 - Now

*Master of Science in Statistics* (GPA: 4.0/4.0)

**Courseworks:** Introduction to Computer Vision, Stochastic Calculus, Learning, Decisions, and Limits

### Tsinghua University

Sept. 2020 - June 2024

*Bachelor of Arts in English* **Outstanding Graduate** (GPA: 3.94/4.0)

*Minor in Economics and Finance* (GPA: 4.0/4.0)

*Minor in Statistics* (without degree) (GPA: 4.0/4.0)

**Courseworks:** Nonparametric Statistics, Linear Regression Analysis, Real Analysis, Econometrics, Applied Stochastic Processes

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## HONOR AND AWARDS

*Academic Excellence Scholarship* (2022-2023)

*Outstanding Graduate of the School of Humanities, Tsinghua University* (2024)

*Outstanding Undergraduate Thesis of the School of Humanities, Tsinghua University* (2024)

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## RESEARCH INTERESTS

Large Language Models, In-context Learning, Interpretability

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## PUBLICATIONS AND MANUSCRIPTS

1 Zhang, Y., Yang, F., Yang, H., & Han, S. (2024). Does checking-in help? Understanding L2 learners' autonomous check-in behavior in an English-language MOOC through learning analytics. *ReCALL*, 1–16. [doi:10.1017/S0958344024000144](https://doi.org/10.1017/S0958344024000144)

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## RESEARCH EXPERIENCES

### Demonstration Selection for In-context learning From a Bayesian Perspective

Oct. 2024 - Now

Advisor: Kaize Ding

- Unite the Bayesian interpretation of ICL and empirical findings in demonstration selection by proposing a multivariate Bayesian characterization of ICL based on LLM hidden states
- Devise methods to improve the ICL performance that can be integrated into any framework of demonstration selection
- Prove that LLM layer updates implicitly performs Bayesian inference

### In-context learning as semantic clustering plus fuzzy copy

July. 2024 - Now

Advisor: Yiqiao Zhong

- Discovered patterns of semantic clustering in the early layer hidden representations of different in-context demonstrations inside LLMs by investigating the (non)linear separability of the representations at different layers
- Provide empirical evidence that LLMs is approximately performing fuzzy copy among the hidden representations of icl prompts by matching the embedding matrices of different attention heads of LLMs
- Propose a two-stage characterization of the ICL mechanisms and validate its existence in various form of NLP and reasoning tasks

### Undergraduate Thesis:

**Automatic Post-Editing of English-Chinese Machine-Translated Academic Texts** Jan. 2024 - June 2024

- Designed and fine-tuned an automatic post-editing model and tested its performance on ParaMed, a English-Chinese parallel corpus that contains academic texts
- Performed comparative statistical analyses as concerning the BLEU and TER scores of the raw machine-translated texts and the automatically post-edited texts
- Qualitatively examined the performance of the model by investigating how it helped to correct the machine translation errors and its failure scenarios

#### **Factor Analysis and Language Education Assessment Instrument**

**Feb. 2023 – June 2023**

- Wrote two chapters on factor analysis in a monograph titled *Statistical Methods in Language Education Studies*, which is currently in press
- Analyzed the similarities and differences between principal component analysis and exploratory factor analysis in theoretical and practical terms and discussed how they tend to be misused in many language education studies
- Applied the two analytical methods to the data collected by administering the Chinese Version of the Foreign Language Enjoyment Scale (C-FLES) to 260 high school English learners in China
- Used the two analytical methods to confirm a three-factor latent construct that underlies the 11 items in the scale and identified two items as being potentially redundant

#### **Check-in Behaviors and Academic Performance in MOOC**

**March 2021 – April 2022**

- Collected and cleaned the academic performance data of 11296 students of an English MOOC on the Xue-TangX platform including their check-in instances and scores in assignments and tests
- Conducted chi-square test of independence and logistic regression to reveal to what extent students' check-in behaviors affect their course completion rates
- Conducted Mann-Whitney U tests between the students who checked in regularly and those who did not check in to determine the differences in their learning behaviors

## **SKILLS**

### **Statistics / Data Analysis:**

Proficiency in R, Python, Stata

### **Languages:**

TOEFL 117: Reading 30, Listening 30, Writing 30, Speaking 27;

GRE 339+5.0: Verbal 169, Quantitative 170, AW 5.0;

Mandarin (native), German (C1)